

THEORY OF COMPUTATION (CS302)

Pumping Lemma for Regular Languages

Learning Objectives

Understand the Pumping Lemma, its conditions, proof concept, applications, and how it is used to prove that languages are not regular.

Introduction

The Pumping Lemma is a fundamental theorem used to prove that certain languages are not regular. It provides a necessary property that every regular language must satisfy.

Formal Statement of Pumping Lemma

For every regular language L , there exists a pumping length p such that any string w in L with length greater than or equal to p can be divided into three parts x , y , and z satisfying specific conditions.

Conditions of Pumping Lemma

1. $w = xyz$
2. $|xy| \leq p$
3. $|y| > 0$
4. $xy^iz \in L$ for all $i \geq 0$

Meaning of the Conditions

The substring y can be repeated any number of times, including zero times, and the resulting string must still belong to the language if the language is regular.

Proof Concept

The proof is based on the finite number of states in a DFA. When a sufficiently long string is processed, some state must repeat, creating a loop that can be pumped.

Steps to Apply Pumping Lemma

- Assume the language is regular.
- Choose a suitable string w .
- Divide w into x , y , and z .
- Pump y by choosing $i = 0$ or $i = 2$.
- Show that the resulting string does not belong to the language.
- Reach a contradiction and conclude the language is not regular.

Solved Example

Language: $L = \{ a^n b^n \mid n \geq 0 \}$

Assume L is regular.

Choose $w = a^n b^n$.

Let y contain only a 's.

Pump y twice.

Result: Number of a 's $\neq$ Number of b 's.

Therefore the new string is not in L.
Contradiction.
Hence L is not regular.

Applications

- Proving languages are not regular
- Automata theory analysis
- Compiler design theory
- Formal language classification

Common Mistakes

- Choosing an invalid string
- Ignoring pumping length constraints
- Not considering all possible decompositions
- Incorrect contradiction arguments

Exam Important Questions

1. State the Pumping Lemma.
2. Explain the conditions of Pumping Lemma.
3. Prove that $a^n b^n$ is not regular.
4. Explain the proof concept of Pumping Lemma.
5. Discuss applications of Pumping Lemma.

Summary

The Pumping Lemma is one of the most important tools in Theory of Computation. It helps determine whether a language is regular by checking if it satisfies the pumping property.